

Backup Types

Lesson 11 — Full, Incremental, Differential and 3-2-1

JKSSB Computer Awareness

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Backup, Storage and Archive: Three Different Jobs

- ▶ **Storage**: where your working data lives every day — the primary copy you actually use.
- ▶ **Backup**: a separate copy kept so data can be recovered after loss, damage, corruption or attack.
- ▶ **Archive**: older data kept for long-term retention, usually removed from routine use.

Exam distinction

A backup is a copy for **recovery**, not just another place to keep files. Storage is the original; archive is old history.

Full Backup — Everything, Every Time

- ▶ Copies **every selected file and folder** (or the whole volume) on every run.
- ▶ + Simplest and fastest to restore: one complete set.
- ▶ - Largest size and longest backup time; repeats unchanged data.

Role

The full backup is your recovery **baseline**. Incremental and differential backups are defined relative to it.

Incremental Backup — Since the Last Backup of Any Type

- ▶ Copies only files changed or added since the **last backup of any type** — full, incremental or differential.
- ▶ + Smallest and quickest to create.
- ▶ - Restore needs the last full backup **plus every** incremental in sequence.

Watch the wording

Incremental looks back only to the immediately preceding backup, whatever type it was.

Differential Backup — Since the Last Full Backup

- ▶ Copies everything changed or added since the **last full backup**.
- ▶ Each differential keeps growing as the day goes on, because it accumulates all changes since that full backup.
- ▶ Restore = the last full backup + the **most recent** differential only.

Role

Differential sits between full and incremental: more data than incremental, but a much shorter restore chain.

Incremental vs Differential: The Near-Neighbour Trap

Type	Copies changes since	Restore needs
Incremental	last backup of any type	full + all incrementals
Differential	last full backup	full + latest differential

The switch word

The word “**full**” is the difference: only the differential always looks back to the last full backup.

Image vs File-Level Backup

Image (system image)

- ▶ A complete clone of a system
- ▶ Operating system, settings, applications and data
- ▶ Restores a whole machine

File-level

- ▶ Selected files and folders only
- ▶ Smaller and more flexible
- ▶ You must reinstall the OS before restoring

An image restores a machine; file-level restores files.

Backup Time vs Restore Time

Type	Backup size / time	Restore speed
Full	Large, slow	Fastest (one set)
Incremental	Smallest, fastest	Slowest (long chain)
Differential	Medium, growing	Medium (short chain)
Image	Largest (whole system)	Whole-machine restore

The trade-off: faster backups usually mean slower restores.

The 3-2-1 Rule

- ▶ **3** copies of your data — the original plus two backups.
- ▶ On **2** different types of media (for example, an external disk and cloud storage).
- ▶ With **1** copy kept off-site (at a different location).

Say it in order

3 copies, 2 media, 1 off-site. Do not swap the numbers.

Offline vs Off-Site, and Cloud Backup

- ▶ **Off-site**: a copy at a different physical location, so one disaster cannot destroy every copy.
- ▶ **Offline**: a copy disconnected from the network — it helps against ransomware that spreads across connected drives.
- ▶ **Cloud backup**: an off-site copy held on remote servers.
- ▶ One copy can be both offline and off-site.

RAID Is Not a Backup

- ▶ RAID gives **redundancy**: if a disk fails, the system keeps running from the mirror or parity.
- ▶ It does **not** protect against accidental deletion, corruption or ransomware — those writes are copied to the other disks too.
- ▶ A deleted file is deleted on every RAID member at once.

RAID $\neq$ backup

Redundancy protects **uptime**; backup protects **recovery**.

Backup Rotation

- ▶ Keep several dated backup sets and reuse them on a fixed schedule (for example, the grandfather–father–son scheme).
- ▶ Rotation keeps an older, **clean** copy available if a problem is only discovered days later.
- ▶ It answers the question: what if today's backup already contains the corruption?

Why rotate

A single overwritten backup gives you no history; rotation keeps recoverable versions across time.

Lesson 11: Recall

- ▶ Full = everything; incremental = changes since the last backup of **any** type; differential = changes since the last **full** backup.
- ▶ Image = whole-system clone; file-level = selected files.
- ▶ 3-2-1 = 3 copies, 2 media, 1 off-site.
- ▶ RAID provides redundancy, not backup — it does not undo deletion, corruption or ransomware.
- ▶ Full restores fastest; incremental restores slowest (longest chain).